

Gitesh Chawda

AI/ML Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years across industry and open source. Author and maintainer of production deep learning libraries spanning transformer architectures, Large Language Model (LLM) inference, quantization, and GPU kernel optimization on PyTorch, TensorFlow, and JAX; 48 merged pull requests into the Hugging Face and Keras ecosystems.

TECHNICAL SKILLS

Languages and Web: Python, C++, TypeScript, JavaScript, SQL, React.js, FastAPI, GraphQL, NumPy, Pandas, OpenCV

ML and Deep Learning: PyTorch, TensorFlow, Keras 3, JAX, Hugging Face Transformers, scikit-learn, Computer Vision, Diffusion Models

LLM Training and Inference: vLLM, TensorRT-LLM, Triton, CUDA, XLA, Quantization (INT8/INT4/FP8), LoRA, PEFT, SFT, GRPO, KV Cache Optimization, FlashAttention, bitsandbytes, Model Serving

Agentic AI: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, OpenAI Agents SDK, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), A2A

MLOps and Cloud: Docker, Linux, Git, GitHub Actions, CI/CD, AWS (Lambda, API Gateway, S3, Cognito), GCP, MongoDB, Neo4j, Chroma

EXPERIENCE

Creator and Maintainer, ZeroModels | Open Source Keras 3 Model Library

Nov 2024 - Present

- Built a backend-agnostic, pip-installable Keras 3 library spanning 100+ vision, speech, LLM, and Vision-Language Model (VLM) architectures (SAM, Whisper, Llama, Qwen) on TensorFlow, PyTorch, and JAX.
- Engineered Hugging Face to Keras checkpoint conversion with bit-close parity validation, and added native weight-only quantization (INT8, INT4, FP8) shrinking model size 8x.
- Grew adoption to 10,000+ monthly PyPI and 100,000+ Hugging Face Hub downloads, supported by automated testing and CI/CD releases on GitHub Actions.

Software Developer, Fiserv | Chennai, India

Jul 2023 - Aug 2024

- Developed and maintained a production banking and payments web application on React.js, GraphQL, and AWS serverless services (AppSync, Lambda, API Gateway, Cognito, S3) powering internal financial workflows.
- Strengthened application security with client-side data tampering prevention and server-side request validation, partnering with AppSec teams to triage and remediate vulnerabilities surfaced in security audits and penetration testing.
- Managed 20+ production deployments and critical hotfixes across development, staging, and production AWS environments, maintaining reliable availability for internal banking and payments workflows.

Open Source Contributor, Hugging Face Transformers, Keras, Keras.io, and KerasCV

Apr 2022 - Present

- Enabled batched multimodal audio generation for Qwen2.5-Omni and Qwen3-Omni, fixed Hunyuan-VL preprocessing parity, and decoupled LXMERT and ConvBERT tokenizers from BertTokenizer in Hugging Face Transformers.
- Shipped Keras 3 multi-backend ops (unravel_index, diagflat, inner, exp2), the PSNR metric API, RandomGrayscale and Equalization layers, the rgb_to_grayscale utility, and a RandomHue fix.
- Extended KerasCV with ConvMixer models, CIoU loss, YOLOv8 box-loss support, and RetinaNet backbone tests, plus official keras.io object detection and super-resolution examples.

Deep Learning Intern, Braynix AI | Bangalore, India

Sep 2021 - Dec 2021

- Designed deep learning pipelines for medical X-ray abnormality detection, raising validation accuracy 20% over baseline Convolutional Neural Network (CNN) models on 5,000+ images.
- Achieved 92% precision and 0.85+ IoU on medical image segmentation using a U-Net architecture with a pretrained EfficientNet-B4 encoder on annotated X-ray data.
- Trained and benchmarked 3 object detection architectures (YOLOv5, SSD, RetinaNet) on 1,000+ labeled medical images, reaching 0.78+ mAP across the evaluated models.

PROJECTS

Trinix: Triton-accelerated PyTorch GPU kernel library with fused attention (MHA, GQA, MQA), RMSNorm, SwiGLU, RoPE, and ALiBi, delivering up to 4.8x faster execution.

InferML: Local-first inference server for 143+ Hugging Face model families (LLM, VLM, diffusion, ASR, TTS) with an OpenAI-compatible API on FastAPI and React.

OpenRAG: Self-hosted RAG framework with switchable vector (Chroma) and knowledge graph (Neo4j) retrieval, cross-encoder reranking, and local LLMs.

txtcode: TypeScript framework orchestrating AI coding agents across 6 messaging platforms and 10 LLM backends.

EDUCATION

Master of Science in Computer Science, California State University, Fullerton, CA

Aug 2024 - May 2026

Bachelor of Engineering in Computer Science, Pimpri Chinchwad College of Engineering | Pune, India

Aug 2019 - Jul 2023

PUBLICATIONS

EffiConvRes: An Efficient CNN with Residual Connections and Depthwise Convolutions. IEEE ICCUBE, 2023.

Survey of Pre-trained CNN and CNN-Transformer Architectures. Smart Grid and Electric Vehicle Conference.